ECE C147/C247, Winter 2026 **Homework #3**
Neural Networks & Deep Learning Prof. J.C. Kao
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Due Friday, 6 Feb 2026, by 11:59pm to Gradescope.
100 points total.

1. (15 points) **Backpropagation for autoencoders.** In an autoencoder, we seek to reconstruct the original data after some operation that reduces the data's dimensionality. We may be interested in reducing the data's dimensionality to gain a more compact representation of the data.

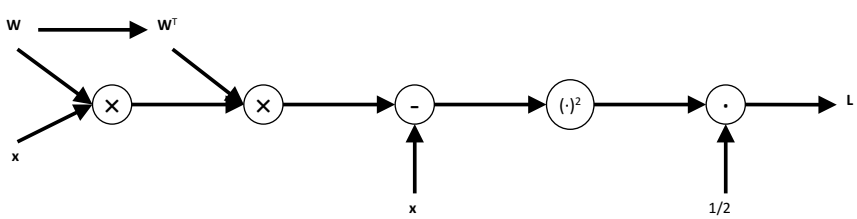
   For example, consider $\mathbf{x} \in \mathbb{R}^n$. Further, consider $\mathbf{W} \in \mathbb{R}^{m \times n}$ where $m < n$. Then $\mathbf{Wx}$ is of lower dimensionality than $\mathbf{x}$. One way to design $\mathbf{W}$ so that it still contains key features of $\mathbf{x}$ is to minimize the following expression

   $$\mathcal{L} = \frac{1}{2} \left\| \mathbf{W}^T \mathbf{W} \mathbf{x} - \mathbf{x} \right\|^2$$

   with respect to $\mathbf{W}$. (To be complete, autoencoders also have a nonlinearity in each layer, i.e., the loss is $\frac{1}{2} \left\| f(\mathbf{W}^T f(\mathbf{Wx})) - x \right\|^2$. However, we'll work with the linear example.)

   (a) (3 points) In words, describe why this minimization finds a $\mathbf{W}$ that ought to preserve information about $\mathbf{x}$.
   Solution: With the loss to be minimized, the difference between output $\mathbf{W}^T\mathbf{Wx}$ and input $\mathbf{x}$ will be minimized. So the hidden representation $\mathbf{Wx}$ will preserve the information about $\mathbf{x}$.

   (b) (3 points) Draw the computational graph for $\mathcal{L}$.
   Solution:



   (c) (3 points) In the computational graph, there should be two paths to $\mathbf{W}$. How do we account for these two paths when calculating $\nabla_{\mathbf{W}} \mathcal{L}$? Your answer should include a mathematical argument.
   Solution: Take a simple example. If the computational graph is composed of two path: $a \to b \to d \to e$ and $a \to c \to d \to e$. In this case, a contributes to e along two paths. Hence, the total derivative of e with respect to as is given by:

   $$\frac{\partial e}{\partial a} = \frac{\partial e}{\partial d} \cdot \frac{\partial d}{\partial b} \cdot \frac{\partial b}{\partial a} + \frac{\partial e}{\partial d} \cdot \frac{\partial d}{\partial c} \cdot \frac{\partial c}{\partial a}$$

(d) (6 points) Calculate the gradient: $\nabla_{\mathbf{W}}\mathcal{L}$.
Solution: Let $\mathbf{K} = \mathbf{W}^T\mathbf{W}\mathbf{x} - \mathbf{x} \in \mathbb{R}^n$ and $\mathbf{P} = \mathbf{W}\mathbf{x} \in \mathbb{R}^m$, so

$$\frac{\partial \mathcal{L}}{\partial \mathbf{K}} = \mathbf{K}$$

Backpropagate to $\mathbf{W}^T$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^T} = \mathbf{K}(\mathbf{W}\mathbf{x})^T$$

Backpropagate to $\mathbf{P}$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{P}} = \mathbf{W}\mathbf{K}$$

Backpropagate to $\mathbf{W}$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \mathbf{W}\mathbf{K}\mathbf{x}^T$$

$\nabla_{\mathbf{W}}\mathcal{L}$ = backprop to $\mathbf{W}$ + backprop to $\mathbf{W}^T$

$$\nabla_{\mathbf{W}}\mathcal{L} = \mathbf{W}\mathbf{K}\mathbf{x}^T + \mathbf{W}\mathbf{x}\mathbf{K}^T$$

2. (15 points) **Backpropagation for Gaussian-process latent variable model.** An important component of unsupervised learning is visualizing high-dimensional data in low-dimensional spaces. One such nonlinear algorithm to do so is from Lawrence, NIPS 2004, called GP-LVM. GP-LVM optimizes the maximum-likelihood of a probabilistic model. We won't get into the details here, but rather to the bottom line: in this paper, a log-likelihood has to be differentiated with respect to a matrix to derive the optimal parameters.

To do so, we will use apply the chain rule for multivariate derivatives via backpropagation. The log-likelihood is:

$$\mathcal{L} = -c - \frac{D}{2}\log|\mathbf{K}| - \frac{1}{2}\mathrm{tr}(\mathbf{K}^{-1}\mathbf{Y}\mathbf{Y}^T)$$

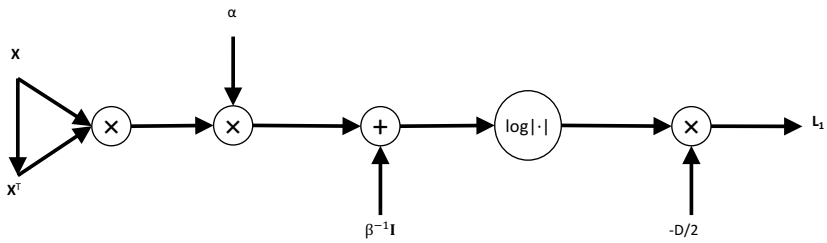
where $\mathbf{K} = \alpha\mathbf{X}\mathbf{X}^T + \beta^{-1}\mathbf{I}$. To solve this, we'll take the derivatives with respect to the two terms with dependencies on $\mathbf{X}$:

$$\begin{aligned}
\mathcal{L}_1 &= -\frac{D}{2}\log|\alpha\mathbf{X}\mathbf{X}^T + \beta^{-1}\mathbf{I}| \\
\mathcal{L}_2 &= -\frac{1}{2}\mathrm{tr}\left((\alpha\mathbf{X}\mathbf{X}^T + \beta^{-1}\mathbf{I})^{-1}\mathbf{Y}\mathbf{Y}^T\right)
\end{aligned}$$

Hint: Lawrence states the gradients in his paper, so you can check your answer. To receive full credit, you will be required to show all work. You may find following equation useful:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{K}} = -\mathbf{K}^{-1}\frac{\partial \mathcal{L}}{\partial \mathbf{K}^{-1}}\mathbf{K}^{-1}$$

(a) (3 points) Draw the computational graph for $\mathcal{L}_1$.
Solution:



(b) (6 points) Compute $\frac{\partial \mathcal{L}_1}{\partial \mathbf{X}}$.
Solution: Calculate the derivative by backpropagation.

$$\mathcal{L}_1 = -\frac{D}{2}\log|\mathbf{K}|, \qquad \mathbf{K} = \alpha\mathbf{X}\mathbf{X}^T + \beta^{-1}\mathbf{I}.$$

Using $\partial \log|\mathbf{K}| = \mathrm{tr}(\mathbf{K}^{-1}\partial\mathbf{K})$, we have

$$\frac{\partial \mathcal{L}_1}{\partial \mathbf{K}} = -\frac{D}{2}\mathbf{K}^{-T}.$$

Since $\mathbf{K}$ depends on $\mathbf{X}$ through $\alpha\mathbf{X}\mathbf{X}^T$,

$$\frac{\partial \mathcal{L}_1}{\partial(\mathbf{X}\mathbf{X}^T)} = \alpha\frac{\partial \mathcal{L}_1}{\partial \mathbf{K}} = -\frac{\alpha D}{2}\mathbf{K}^{-T}.$$

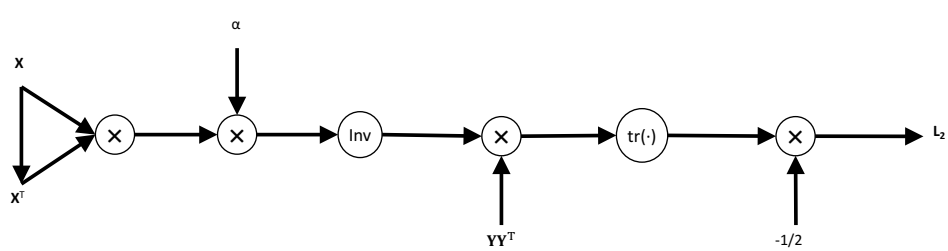
Now use $\partial(\mathbf{X}\mathbf{X}^T) = \partial\mathbf{X}\,\mathbf{X}^T + \mathbf{X}\,\partial\mathbf{X}^T$ to obtain

$$\frac{\partial \mathcal{L}_1}{\partial \mathbf{X}} = \left(\frac{\partial \mathcal{L}_1}{\partial(\mathbf{X}\mathbf{X}^T)} + \left(\frac{\partial \mathcal{L}_1}{\partial(\mathbf{X}\mathbf{X}^T)}\right)^T\right)\mathbf{X} = -\alpha D\,(\mathbf{K}^{-T})\,\mathbf{X}.$$

Because $\mathbf{K}$ is symmetric ($\mathbf{K}^T = \mathbf{K}$), $\mathbf{K}^{-T} = \mathbf{K}^{-1}$ and $(\mathbf{K}^{-T}) = \mathbf{K}^{-1}$, hence

$$\boxed{\frac{\partial \mathcal{L}_1}{\partial \mathbf{X}} = -\alpha D\,\mathbf{K}^{-1}\mathbf{X}}$$

(c) (3 points) Draw the computational graph for $\mathcal{L}_2$.
Solution:



(d) (6 points) Compute $\frac{\partial \mathcal{L}_2}{\partial \mathbf{X}}$.
Solution: Calculate the derivative by backpropagation.

$$\mathcal{L}_2 = -\frac{1}{2}\mathrm{tr}(\mathbf{K}^{-1}\mathbf{Y}\mathbf{Y}^T).$$

Using $\frac{\partial}{\partial \mathbf{A}} \mathrm{tr}(\mathbf{AB}) = \mathbf{B}^T$,

$$\frac{\partial \mathcal{L}_2}{\partial \mathbf{K}^{-1}} = -\frac{1}{2}(\mathbf{Y}\mathbf{Y}^T)^T = -\frac{1}{2}\mathbf{Y}\mathbf{Y}^T.$$

Apply the identity (given in the prompt)

$$\frac{\partial \mathcal{L}}{\partial \mathbf{K}} = -\mathbf{K}^{-1}\frac{\partial \mathcal{L}}{\partial \mathbf{K}^{-1}}\mathbf{K}^{-1},$$

yielding

$$\frac{\partial \mathcal{L}_2}{\partial \mathbf{K}} = -\mathbf{K}^{-1}\left(-\frac{1}{2}\mathbf{Y}\mathbf{Y}^T\right)\mathbf{K}^{-1} = \frac{1}{2}\mathbf{K}^{-1}\mathbf{Y}\mathbf{Y}^T\mathbf{K}^{-1}.$$

Since $\mathbf{K} = \alpha\mathbf{X}\mathbf{X}^T + \beta^{-1}\mathbf{I}$,

$$\frac{\partial \mathcal{L}_2}{\partial(\mathbf{X}\mathbf{X}^T)} = \alpha\frac{\partial \mathcal{L}_2}{\partial \mathbf{K}} = \frac{\alpha}{2}\mathbf{K}^{-1}\mathbf{Y}\mathbf{Y}^T\mathbf{K}^{-1}.$$

Using the same symmetrization step as above,

$$\frac{\partial \mathcal{L}_2}{\partial \mathbf{X}} = \left(\frac{\partial \mathcal{L}_2}{\partial(\mathbf{X}\mathbf{X}^T)} + \left(\frac{\partial \mathcal{L}_2}{\partial(\mathbf{X}\mathbf{X}^T)}\right)^T\right)\mathbf{X}.$$

Here $\mathbf{K}^{-1}$ and $\mathbf{Y}\mathbf{Y}^T$ are symmetric, so the matrix is symmetric and we get

$$\boxed{\frac{\partial \mathcal{L}_2}{\partial \mathbf{X}} = \alpha\,\mathbf{K}^{-1}\mathbf{Y}\mathbf{Y}^T\mathbf{K}^{-1}\mathbf{X}}$$

(e) (2 points) Compute $\frac{\partial \mathcal{L}}{\partial \mathbf{X}}$.
Solution: Since $\mathcal{L} = -c + \mathcal{L}_1 + \mathcal{L}_2$ and c does not depend on $\mathbf{X}$,

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}} = \frac{\partial \mathcal{L}_1}{\partial \mathbf{X}} + \frac{\partial \mathcal{L}_2}{\partial \mathbf{X}}.$$

Substituting results from parts (b) and (d),

$$\boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{X}} = \alpha\,\mathbf{K}^{-1}\mathbf{Y}\mathbf{Y}^T\mathbf{K}^{-1}\mathbf{X} \;-\; \alpha D\,\mathbf{K}^{-1}\mathbf{X}}$$

3. (20 points, Optional for C147/Required for C247) **Backpropagation through a Residual Block with Biases.**

Training very deep neural networks is notoriously difficult due to problems such as vanishing gradients and unstable optimization. A key architectural idea that enables the successful training of modern deep models—including residual networks, Transformers, and large language models (LLMs)—is the *residual (skip) connection.*

In a standard feedforward layer, an input $\mathbf{x}$ is transformed into an output $F(\mathbf{x})$. A residual connection modifies this computation by explicitly adding the input to the output:

$$\mathbf{x} \longrightarrow F(\mathbf{x}) + \mathbf{x}.$$

Rather than learning an entirely new representation at each layer, the model learns a *residual function*—a correction to the input—which significantly improves gradient flow during training.

In this problem, you will analyze backpropagation through a simple residual block that includes both weight matrices and bias terms, and explicitly see how the skip connection alters gradient propagation.

Consider the following network:

$$\mathbf{h} = \sigma(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1), \qquad \hat{\mathbf{y}} = \mathbf{W}_2\mathbf{h} + \mathbf{b}_2 + \mathbf{x},$$

where:

- $\mathbf{x} \in \mathbb{R}^d$ is the input,
- $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{d\times d}$ are trainable weight matrices,
- $\mathbf{b}_1, \mathbf{b}_2 \in \mathbb{R}^d$ are trainable bias vectors,
- $\sigma(\cdot)$ denotes the element-wise sigmoid activation function,
- the term $+\mathbf{x}$ represents a residual (identity) connection.

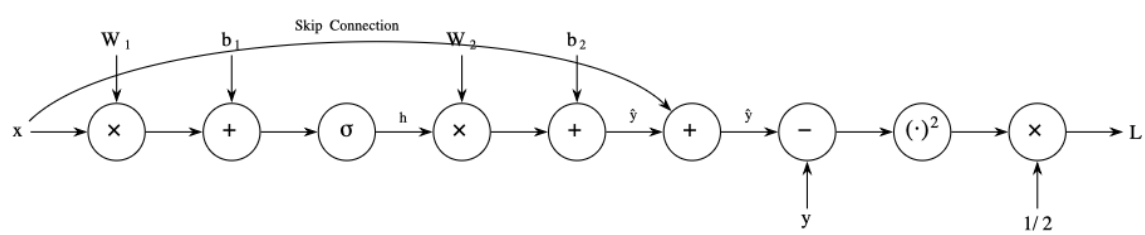
The loss function is the squared error:

$$\mathcal{L} = \frac{1}{2}\|\hat{\mathbf{y}} - \mathbf{y}\|^2.$$

(a) (4 points) Draw the **computational graph** corresponding to this model and loss. Your graph should clearly indicate:

- the affine transformation and nonlinearity in the main branch,
- the identity (skip) connection,
- the addition operation where the two paths merge,
- the role of bias terms in each layer.

Solution:



(b) (5 points) Compute the gradient of the loss with respect to $\mathbf{W}_2$, i.e.,

$$\nabla_{\mathbf{W}_2}\mathcal{L}.$$

Express your answer using matrix/vector notation and clearly show how the error signal propagates from the loss to the output weights.

Solution: Define the output error (loss-to-output gradient)

$$\boldsymbol{\delta} \equiv \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} = \hat{\mathbf{y}} - \mathbf{y}.$$

Since $\hat{\mathbf{y}} = \mathbf{W}_2\mathbf{h} + \mathbf{b}_2 + \mathbf{x}$, we have

$$\nabla_{\mathbf{W}_2}\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}}\frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{W}_2} = \boldsymbol{\delta}\,\mathbf{h}^T.$$

(c) (3 points) Compute the gradient of the loss with respect to $\mathbf{b}_2$, i.e.,

$$\nabla_{\mathbf{b}_2}\mathcal{L}.$$

Briefly explain why the residual connection does or does not affect this gradient.

Solution: Because $\mathbf{b}_2$ enters additively,

$$\nabla_{\mathbf{b}_2}\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}}\frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{b}_2} = \boldsymbol{\delta} \cdot \mathbf{I} = \boldsymbol{\delta}.$$

The skip connection bypasses $\mathbf{b}_2$, so it does not change $\partial\hat{\mathbf{y}}/\partial\mathbf{b}_2$.

(d) (5 points) Compute the gradient of the loss with respect to $\mathbf{W}_1$, i.e.,

$$\nabla_{\mathbf{W}_1}\mathcal{L}.$$

In your derivation:

- explicitly apply the chain rule,
- show where the derivative of the sigmoid function appears,
- make clear how the residual connection contributes to the gradient signal reaching $\mathbf{W}_1$.

Solution: First note

$$\frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{h}} = \mathbf{W}_2, \qquad \frac{\partial \mathbf{h}}{\partial \mathbf{z}_1} = \mathrm{diag}(\sigma'(\mathbf{z}_1)), \qquad \frac{\partial \mathbf{z}_1}{\partial \mathbf{W}_1} = \mathbf{x}^T.$$

By the chain rule,

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}} = \left(\frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{h}}\right)^T \frac{\partial \mathcal{L}}{\partial \hat{\mathbf{y}}} = \mathbf{W}_2^T\boldsymbol{\delta}.$$

Then

$$\frac{\partial \mathcal{L}}{\partial \mathbf{z}_1} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}} \odot \sigma'(\mathbf{z}_1) = \left(\mathbf{W}_2^T\boldsymbol{\delta}\right) \odot \sigma'(\mathbf{z}_1),$$

and finally

$$\nabla_{\mathbf{W}_1}\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \mathbf{z}_1}\frac{\partial \mathbf{z}_1}{\partial \mathbf{W}_1} = \left(\mathbf{W}_2^T\boldsymbol{\delta} \odot \sigma'(\mathbf{z}_1)\right)\mathbf{x}^T.$$

The residual connection is what makes $\boldsymbol{\delta} = \partial\mathcal{L}/\partial\hat{\mathbf{y}}$

(e) (3 points) Compute the gradient of the loss with respect to $\mathbf{b}_1$, i.e.,

$$\nabla_{\mathbf{b}_1} \mathcal{L}.$$

Solution: Since $\mathbf{z}_1 = \mathbf{W}_1 \mathbf{x} + \mathbf{b}_1$,

$$\nabla_{\mathbf{b}_1} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \mathbf{z}_1} \frac{\partial \mathbf{z}_1}{\partial \mathbf{b}_1} = \left(\mathbf{W}_2^T \boldsymbol{\delta} \odot \sigma'(\mathbf{z}_1)\right) \cdot \mathbf{I} = \mathbf{W}_2^T \boldsymbol{\delta} \odot \sigma'(\mathbf{z}_1).$$

4. (30 points) **2-layer neural network.** Compete the two-layer neural network Jupyter notebook. Print out the entire workbook and relevant code and submit it as a pdf to gradescope.

5. (20 points) **General FC neural network.** Compete the FC Net Jupyter notebook. Print out the entire workbook and relevant code and submit it as a pdf to gradescope.